

# torch.nn.modules.module.register\_module

[https://pytorch.org/docs/stable/generated/torch.nn.modules.module.register\\_m](https://pytorch.org/docs/stable/generated/torch.nn.modules.module.register_m)

## Description:

Register a global forward hook for all the modules. This adds global state to the nn.module module and it is only intended for debugging/profiling purposes. The hook will be called every time after forward() has computed an output. It should have the following signature: The input contains only the positional arguments given to the module. Keyword arguments won't be passed to the hooks and only to the forward. You can optionally modify the output of the module by returning a new value that will replace the output from the forward() function.

## Function Signature

---

```
torch.nn.modules.module.register_module_forward_hook(hook,  
*, with_kwargs=False, always_call=False)[source]#
```

Parameters

Returns

Return type

### Returns:

`torch.utils.hooks.RemovableHandle`

## Code Examples

---

```
hook(module, input, output) -> None or modified output
```

## Notes & Warnings

---

### WARNING

Warning This adds global state to the `nn.module` module and it is only intended for debugging/profiling purposes.

# Detailed Documentation

---

## Documentation

Register a global forward hook for all the modules.

This adds global state to the `nn.module` module and it is only intended for debugging/profiling purposes.

The hook will be called every time after `forward()` has computed an output. It should have the following signature:

The input contains only the positional arguments given to the module. Keyword arguments won't be passed to the hooks and only to the `forward`. You can optionally modify the output of the module by returning a new value that will replace the output from the `forward()` function.

`hook (Callable)` – The user defined hook to be registered.

`always_call (bool)` – If `True` the hook will be run regardless of whether an exception is raised while calling the Module. Default: `False`

a handle that can be used to remove the added hook by calling `handle.remove()`

`torch.utils.hooks.RemovableHandle`

This hook will be executed before specific module hooks registered with `register_forward_hook`.

